Sardar Vallabhbhai National Institute of Technology (SVNIT), Surat

Department of Artificial Intelligence

Agentic AI (AI465), Semester VII

Research Project — Phase 1

Paper Study, Identification of Research Gaps and Formulation of an Objective

Maximum marks: 20 | Individual work | Presentation with a short written submission | This component replaces the quiz.

1. Purpose

The laboratory experiments teach you to build agents that are already known to work. This component asks the opposite question: given the published work on a topic of this course, what is not yet solved, and what is one thing you could do about it? Agentic AI is a field where results are new, benchmarks are unsettled and claims are often made without reporting cost or variance, so there is a great deal to find. Phase 1 ends with a single, defensible objective; Phase 2 asks you to implement that same objective. The component addresses course outcomes CO3, CO4 and CO5 and carries 20 marks in place of the quiz.

2. Working rules and topic

1. The project is individual. Every student selects their own papers, writes their own summary and presents alone. Discussing ideas with classmates is allowed; sharing text, slides or a set of gaps is not, and identical submissions will be treated as copying by all students involved.
2. The topic must fall inside the AI465 syllabus. Choose one of the areas below and register your primary paper before you begin, so that no two students in the section work on the same paper.

Syllabus unit	Example project areas
Foundations and agent architectures	autonomy levels, BDI mapped onto LLM agents, when a workflow beats an autonomous agent
Decision making and planning	MDP and POMDP planning, belief-state updates, LLM planners compared with classical planners
LLMs as reasoning engines	chain-of-thought and self-consistency, reflection and self-critique, plan-then-execute, ReAct loops
Agentic workflows and frameworks	planner–executor–critic separation, workflow graphs, retry and fallback, error propagation, trace debugging
Tool use and MCP	schema-based function calling, tool selection and routing, failure and partial completion, sandboxing and permission control
Memory, retrieval and context	RAG pipelines, chunking and indexing strategies, episodic memory, summarisation and compression, session persistence
Multi-agent LLM systems	debate and deliberation, supervisor–worker delegation, negotiation, orchestration and scalability
Evaluation, safety and governance	defining task success, LLM judges, regression and adversarial testing, prompt injection, cost and latency modelling

3. Choosing the papers

1. Study two to three papers on one narrow problem, not three papers on three unrelated problems. One of them is your primary paper; the others are read to place it in context.
2. The primary paper must be from 2024 or later, because work older than that is usually built on models that are no longer available. Preferred venues are NeurIPS, ICLR, ICML, ACL, EMNLP, NAACL, COLM, AAAI and AAMAS.
3. Much of this field appears on arXiv first. A preprint may be your primary paper only if it has a public implementation and has been cited by other groups; say so explicitly when you register it.
4. At most one survey may be used, and only for context. A survey cannot be the primary paper, because gaps reported by a survey are the survey author's findings, not yours.
5. The primary paper must report experiments on a named benchmark or task suite. You cannot identify a gap in an evaluation that does not exist.

4. What counts as a research gap

A research gap is a specific limitation you can point to in the paper — in its method, its assumptions or its evaluation — together with a reason why it matters. Identify three or four gaps, each supported by a section, table or figure of the paper. The categories below are a checklist for reading, not a list to fill in mechanically; a good submission usually finds gaps of at least two different kinds.

Kind of gap	The question to ask while reading
Assumption	What must the model, the tools or the environment provide for this to work, and where would that fail?
Cost and latency	How many model calls and tokens does one task consume, and is the gain worth it against a simpler method given the same budget?
Evaluation	Is the benchmark saturated or possibly in the training data, is the judge independent of the system it grades, are results averaged over enough runs?
Generality	Was it tested on one model family or one domain only, and what breaks on a second one?
Reliability	How does the agent behave when a tool fails, a plan is wrong, or the context window overflows?
Reproducibility	Are the prompts, the model version and the seeds given, and would the result survive that model being updated?

The following do not earn marks: copying the paper's own future-work sentences without analysis; vague complaints such as accuracy could be improved; asking for a bigger model with no argument; and any gap already closed by a paper published since.

5. Formulating the objective

From your gaps, derive exactly one objective, and choose the gap you can actually address in the time available rather than the most important one. The objective must be a single sentence naming what you will build, what you will compare it against and how you will measure the difference. Use this template:

```
To <build / extend / evaluate> <specific agent component or mechanism>
   for <task> on <named benchmark or task suite, with its size stated>,
   compared against <named baseline> at a matched budget,
   measured by <task metric> and <cost metric>.
```

An objective is feasible if the benchmark is public and you can work on a subset of one or two hundred tasks, the models are reachable through a free tier or run locally through an open-weight model, and one Colab session or a laptop suffices. Budget the token cost before you commit: estimate calls per task multiplied by tasks multiplied by runs, and if the answer needs a paid account it is not feasible. Nothing requiring model training or fine-tuning at scale will be approved.

Worked example

Primary paper	A 2024 paper on multi-agent debate for factual question answering: three agents argue for three rounds and a judge model picks the final answer, evaluated on a 500-question subset of a public QA benchmark.
Gaps	(i) the method costs roughly nine model calls per question against one for the baseline, and no token or latency figure is reported anywhere in the paper; (ii) the judge is from the same model family as the debaters, so self-preference is untested; (iii) every configuration is run once although the decoding temperature is 0.7; (iv) only one model family is used, so it is unknown whether the gain comes from debate or from that model being strong at self-critique.
Objective	To evaluate two-agent debate with an independent open-weight judge on the same 500-question subset, compared against single-agent self-consistency at a matched token budget, measured by exact-match accuracy and by tokens and wall-clock latency per question.

6. Deliverables

1. A written summary of at most two pages containing: full citations of the papers read, half a page on the primary paper's problem and method in your own words, the three or four gaps with the evidence for each, and the objective sentence.
2. A slide deck of six to eight slides following the same order.
3. A presentation of six minutes with three minutes of questions.
4. Submit as RP1_Roll_No.pdf before the presentation slot.

The summary must be in your own words. Text copied from the papers, from a web page or from a language model and presented as your own analysis will be treated as plagiarism and scored zero for the component. You may use language models to find, translate and summarise papers for yourself; say so in one line at the end of the summary, and be ready to explain any sentence in it during questions. This course studies these systems, so using them is expected — passing their output off as your analysis is not.

7. Evaluation

Criterion	What earns full marks	Marks
Choice of papers	recent, on one narrow problem, clearly within the syllabus, venue rules respected	3
Understanding of the primary paper	problem, method and reported results explained in your own words; questions on the agent loop, the prompts and the evaluation answered correctly	5
Research gaps	three or four gaps, each pointing to specific evidence in the paper, of at least two different kinds, none of them already closed in the literature	6
Objective	one sentence, specific, measurable, feasible within the token budget, and traceable to one of the gaps identified	4
Presentation and questions	time kept, slides readable, questions answered without reading from the slides	2

8. Schedule

Week	Milestone
1	area chosen and primary paper registered
2	papers approved; primary paper read and its evaluation setup understood
3	gaps drafted and discussed in the lab slot; token budget estimated
4	Phase 1 presentation and submission

The objective approved at the end of Phase 1 is the one you implement in Phase 2. It may be changed only with prior approval, and only for a demonstrated technical reason.